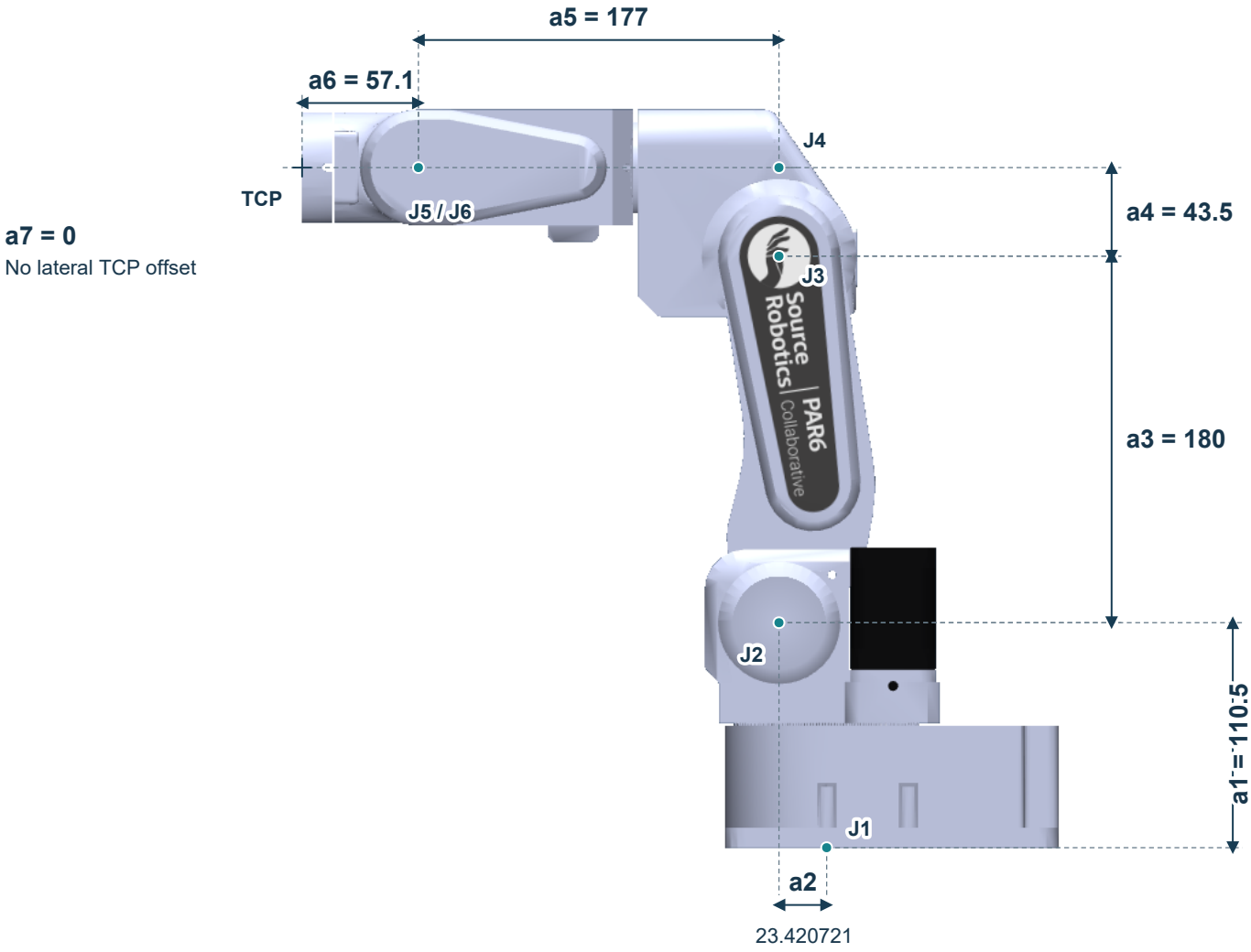
PAR6 / Flange only

Kinematic review | Standard DH to the configured TCP

Pose: q1 ... q6 = 0 rad; no jaws. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.420721	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = 0	-a6 = -57.1

TCP is the configured flange reference. No gripper is installed.
a6 measures J6 center to flange; a7 is zero.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.004048 mm / 0.001350 deg
Max ideal DH vs URDF: 0.000000 mm / 0.000000 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

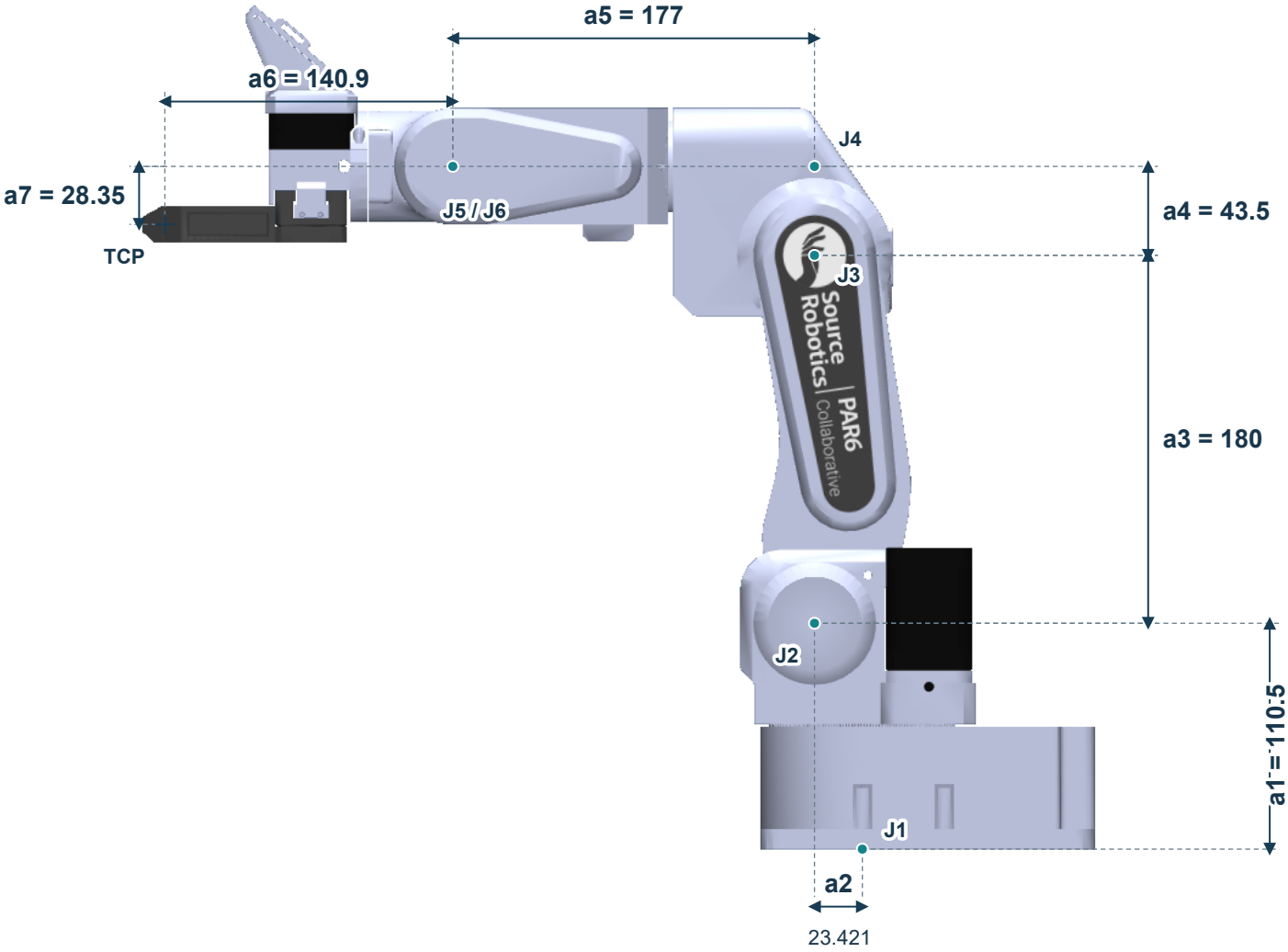
Source: par6_flange.urdf + scene_flange.xml | Full transforms, hashes and checks: flange.json

All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

PAR6 / MSG / Solid jaws

Kinematic review | Standard DH to the configured TCP

Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.421	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = -28.35	-a6 = -140.9

MSG jaw variants share this operational TCP; it is not a measured grasp center.
The 28.35 mm lateral TCP offset is included in DH row 6 as a = -a7.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.001285 mm / 0.000547 deg
Max ideal DH vs URDF: 0.004992 mm / 0.000860 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

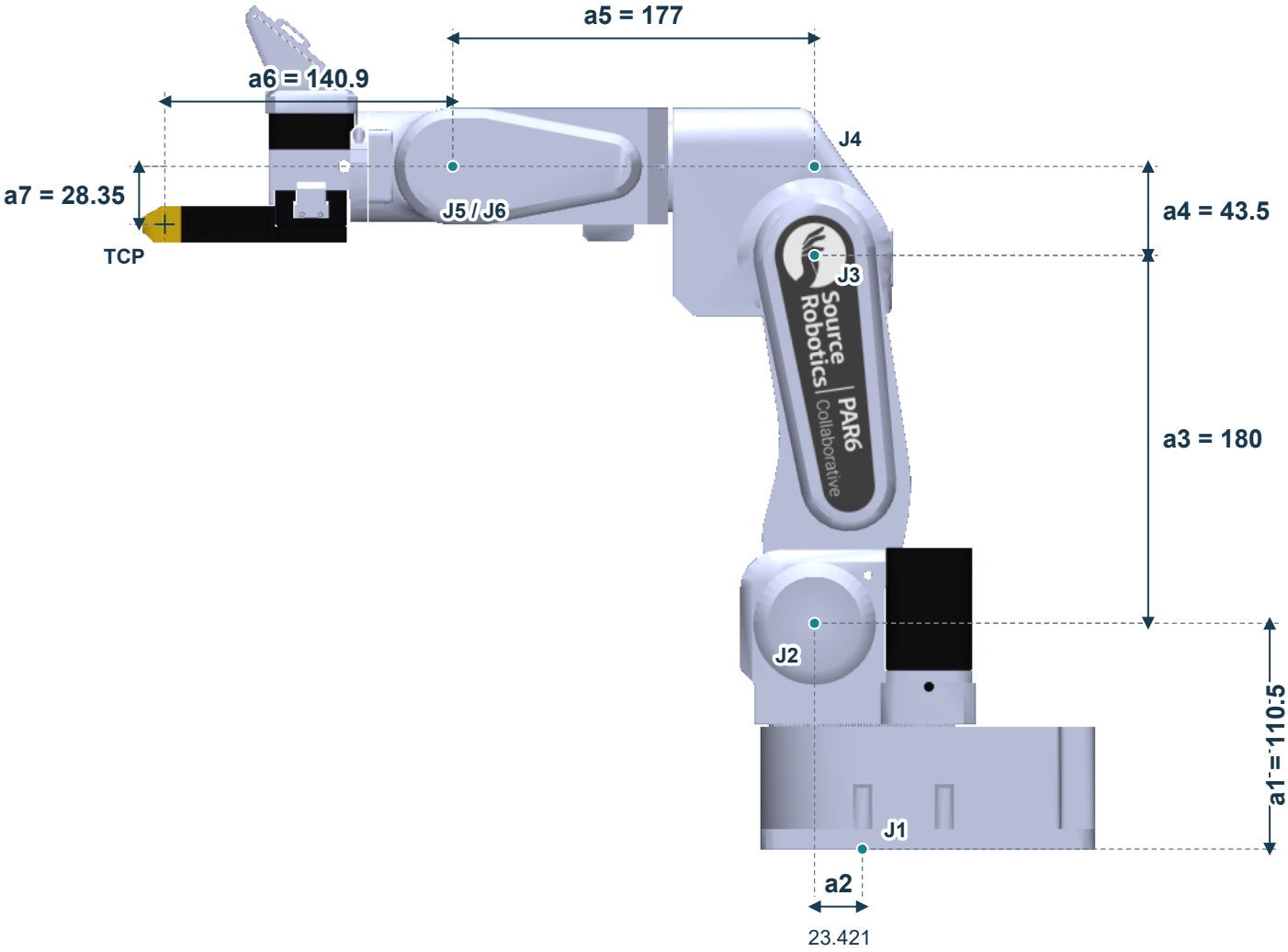
Source: par6_msg.urdf + scene_msg.xml | Full transforms, hashes and checks: msg.json

All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

PAR6 / MSG / Soft TPU jaws

Kinematic review | Standard DH to the configured TCP

Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.421	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = -28.35	-a6 = -140.9

MSG jaw variants share this operational TCP; it is not a measured grasp center.
The 28.35 mm lateral TCP offset is included in DH row 6 as a = -a7.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.001285 mm / 0.000547 deg
Max ideal DH vs URDF: 0.004992 mm / 0.000860 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

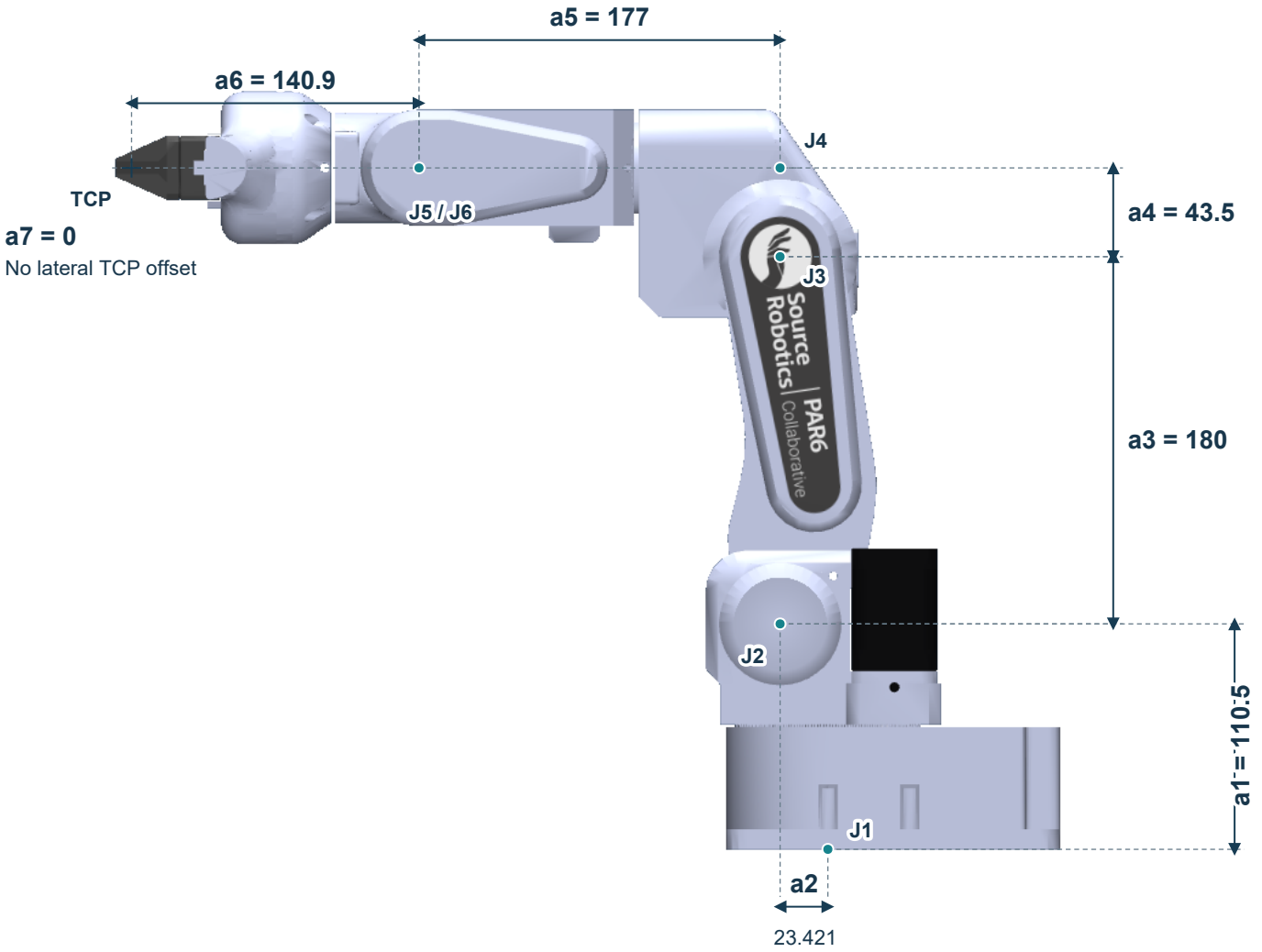
Source: par6_msg_tpu_glue.urdf + scene_msg_tpu_glue.xml | Full transforms, hashes and checks: msg_tpu_glue.json

All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

PAR6 / SSG48 / Solid jaws

Kinematic review | Standard DH to the configured TCP

Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.421	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = 0	-a6 = -140.9

SSG48 jaw variants share this operational TCP; it is not a measured grasp center.
Solid jaws are about 23.4 mm shorter than Fin-ray jaws; their TCP was retained.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.000915 mm / 0.000339 deg
Max ideal DH vs URDF: 0.005262 mm / 0.001048 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

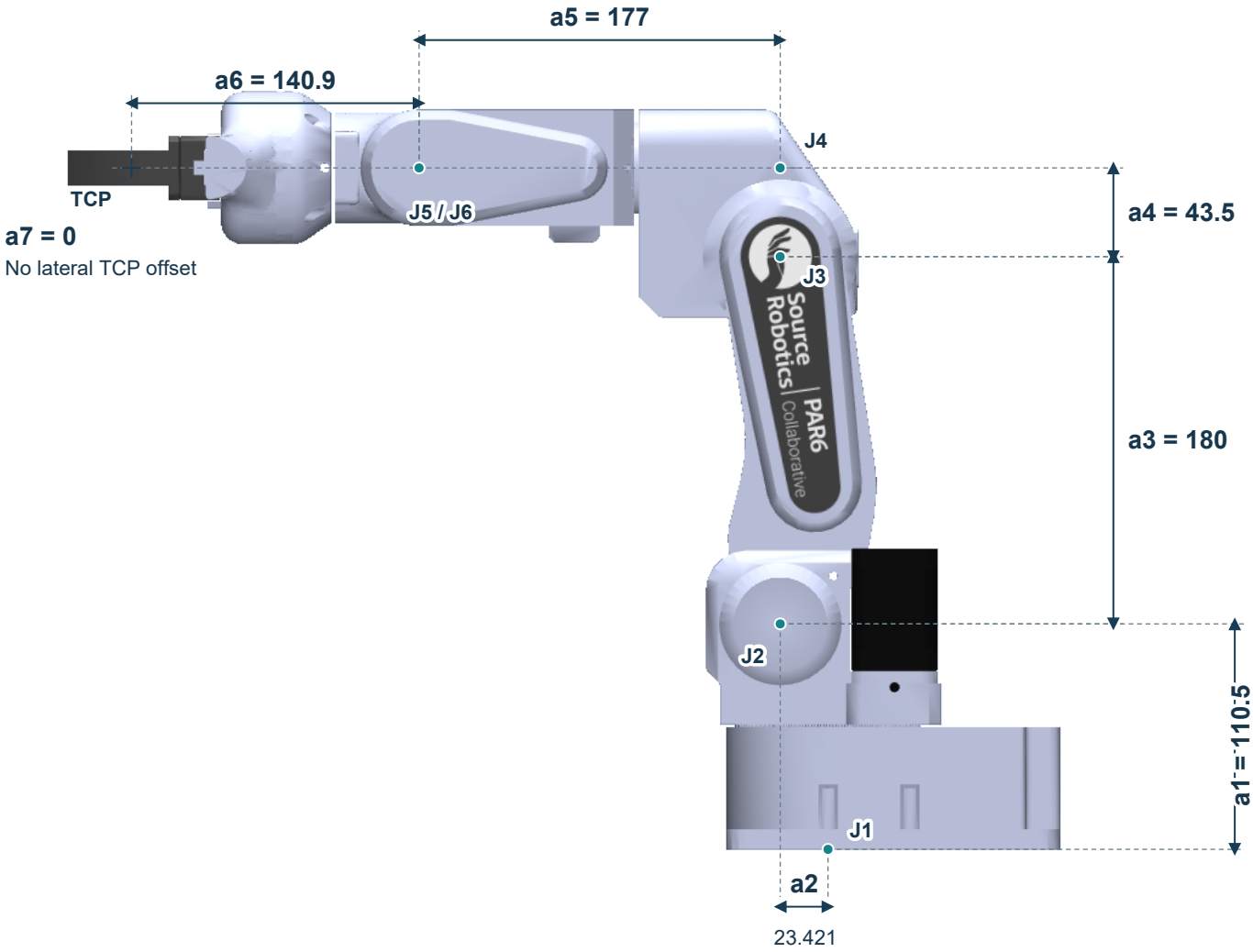
Source: par6_ssg48.urdf + scene_ssg48.xml | Full transforms, hashes and checks: ssg48.json

All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

PAR6 / SSG48 / Fin-ray jaws

Kinematic review | Standard DH to the configured TCP

Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.421	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = 0	-a6 = -140.9

SSG48 jaw variants share this operational TCP; it is not a measured grasp center.
Solid jaws are about 23.4 mm shorter than Fin-ray jaws; their TCP was retained.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.000915 mm / 0.000339 deg
Max ideal DH vs URDF: 0.005262 mm / 0.001048 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

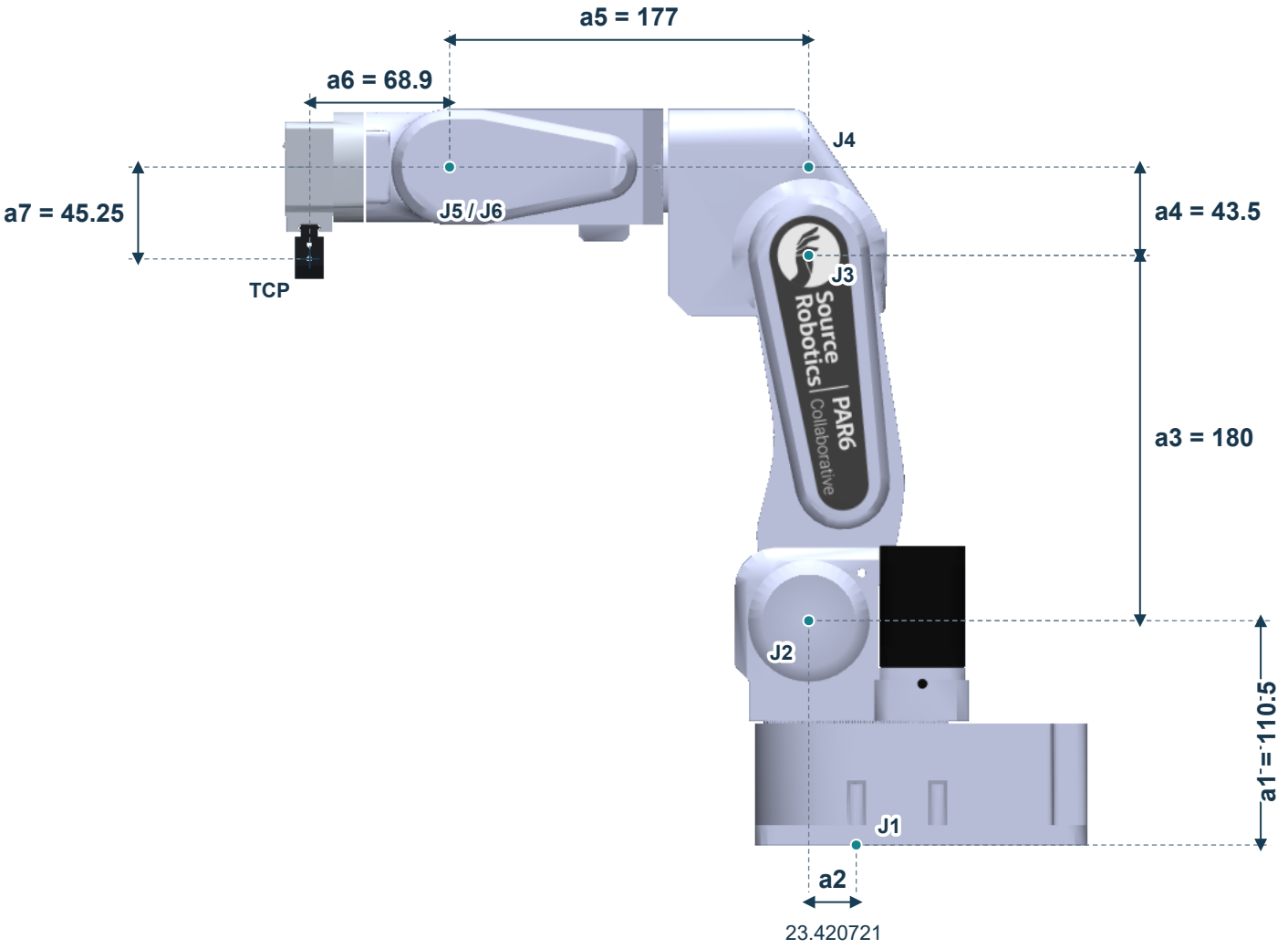
Source: par6_ssg48_stock.urdf + scene_ssg48_stock.xml | Full transforms, hashes and checks: ssg48_stock.json

All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

PAR6 / Pneumatic gripper

Kinematic review | Standard DH to the configured TCP

Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.420721	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = -45.25	-a6 = -68.9

Pneumatic TCP follows the legacy MHZ2-10D reference; it is not a newly measured grasp center.
Jaws close symmetrically through 3.5 mm each. Mass and finite actuator settings are estimates.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.004340 mm / 0.001350 deg
Max ideal DH vs URDF: 0.000000 mm / 0.000000 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

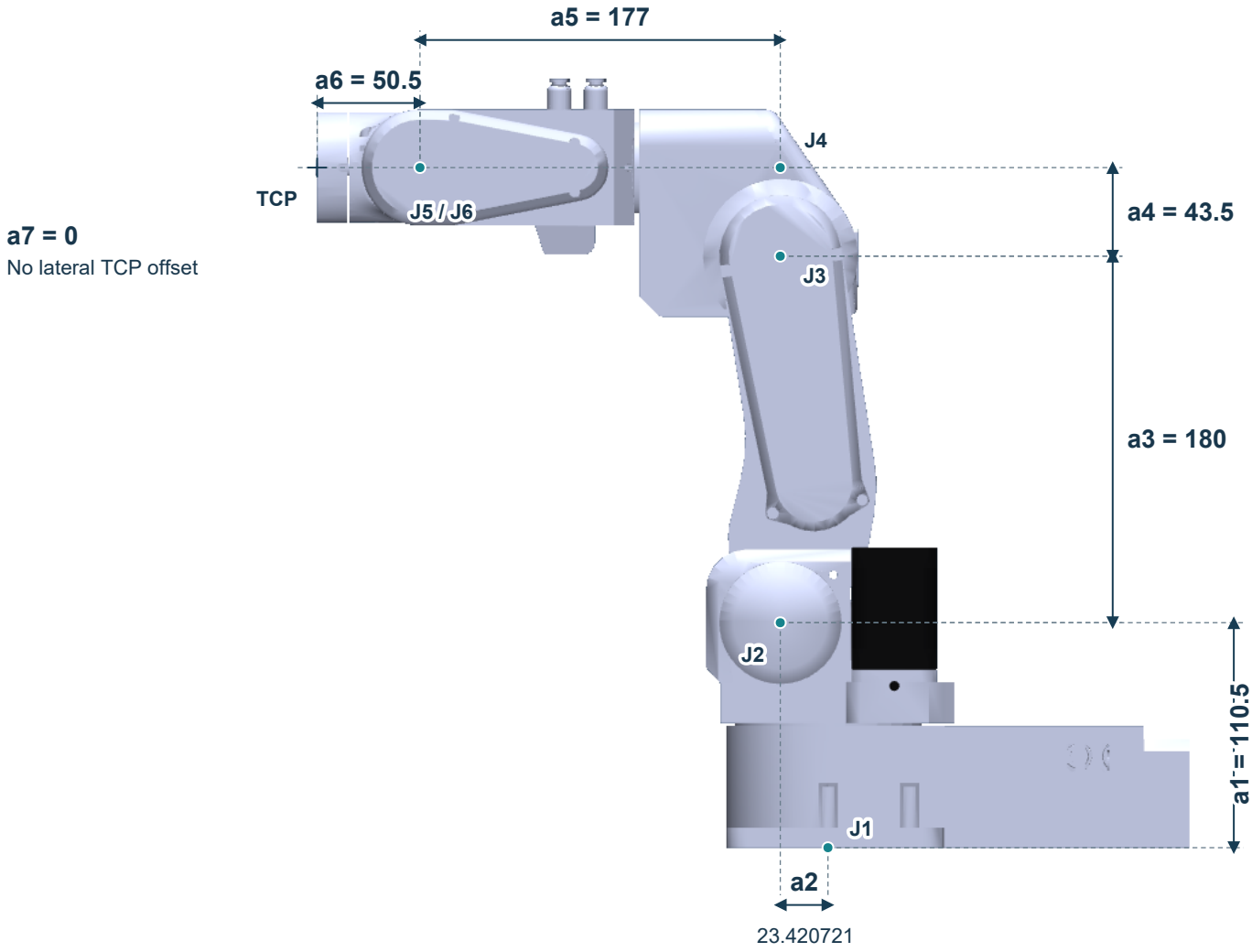
Source: par6_pneumatic.urdf + scene_pneumatic.xml | Full transforms, hashes and checks: pneumatic.json

All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

PAROL6 / Flange only

Kinematic review | Standard DH to the configured TCP

Pose: q1 ... q6 = 0 rad; no jaws. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.420721	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = 0	-a6 = -50.5

TCP is the configured flange reference. No gripper is installed.
a6 measures J6 center to flange; a7 is zero.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.002206 mm / 0.001026 deg
Max ideal DH vs URDF: 0.000000 mm / 0.000000 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

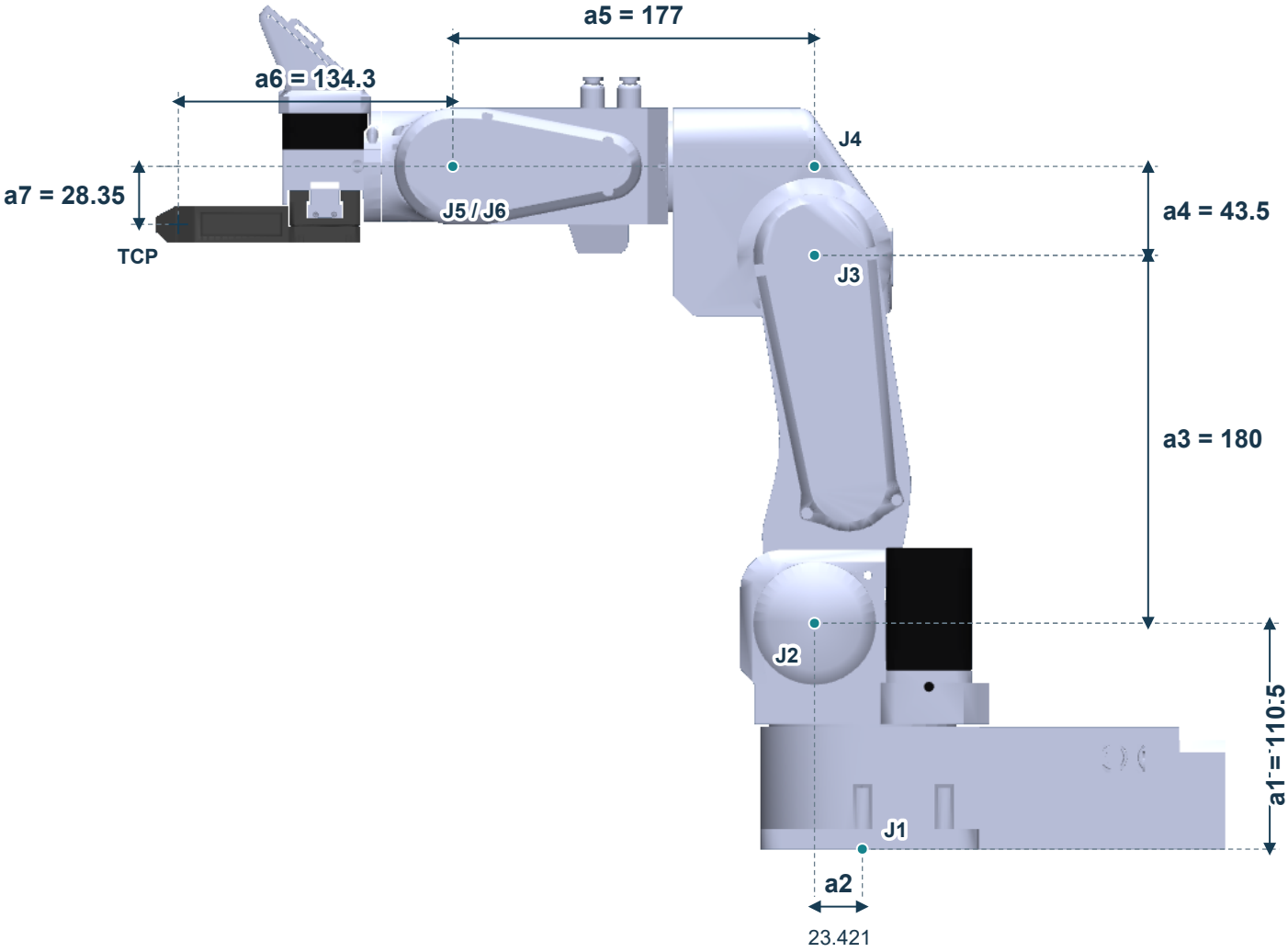
Source: parol6_flange.urdf + scene_flange.xml | Full transforms, hashes and checks: flange.json

All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

PAROL6 / MSG / Solid jaws

Kinematic review | Standard DH to the configured TCP

Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.421	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = -28.35	-a6 = -134.3

MSG jaw variants share this operational TCP; it is not a measured grasp center.
The 28.35 mm lateral TCP offset is included in DH row 6 as a = -a7.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.001000 mm / 0.000479 deg
Max ideal DH vs URDF: 0.002787 mm / 0.000679 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

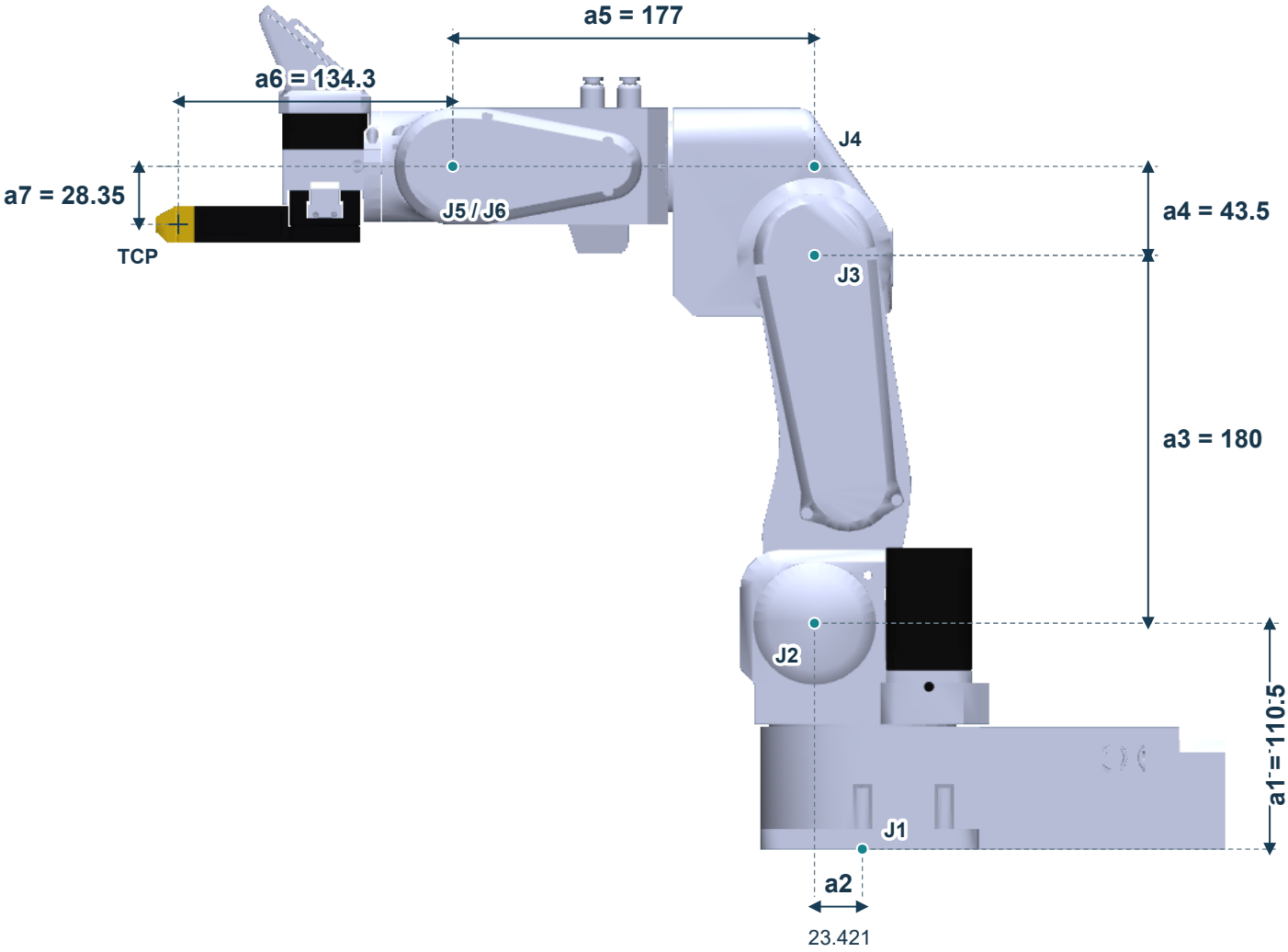
Source: parol6_msg.urdf + scene_msg.xml | Full transforms, hashes and checks: msg.json

All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

PAROL6 / MSG / Soft TPU jaws

Kinematic review | Standard DH to the configured TCP

Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.421	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = -28.35	-a6 = -134.3

MSG jaw variants share this operational TCP; it is not a measured grasp center.
The 28.35 mm lateral TCP offset is included in DH row 6 as a = -a7.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.001000 mm / 0.000479 deg
Max ideal DH vs URDF: 0.002787 mm / 0.000679 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

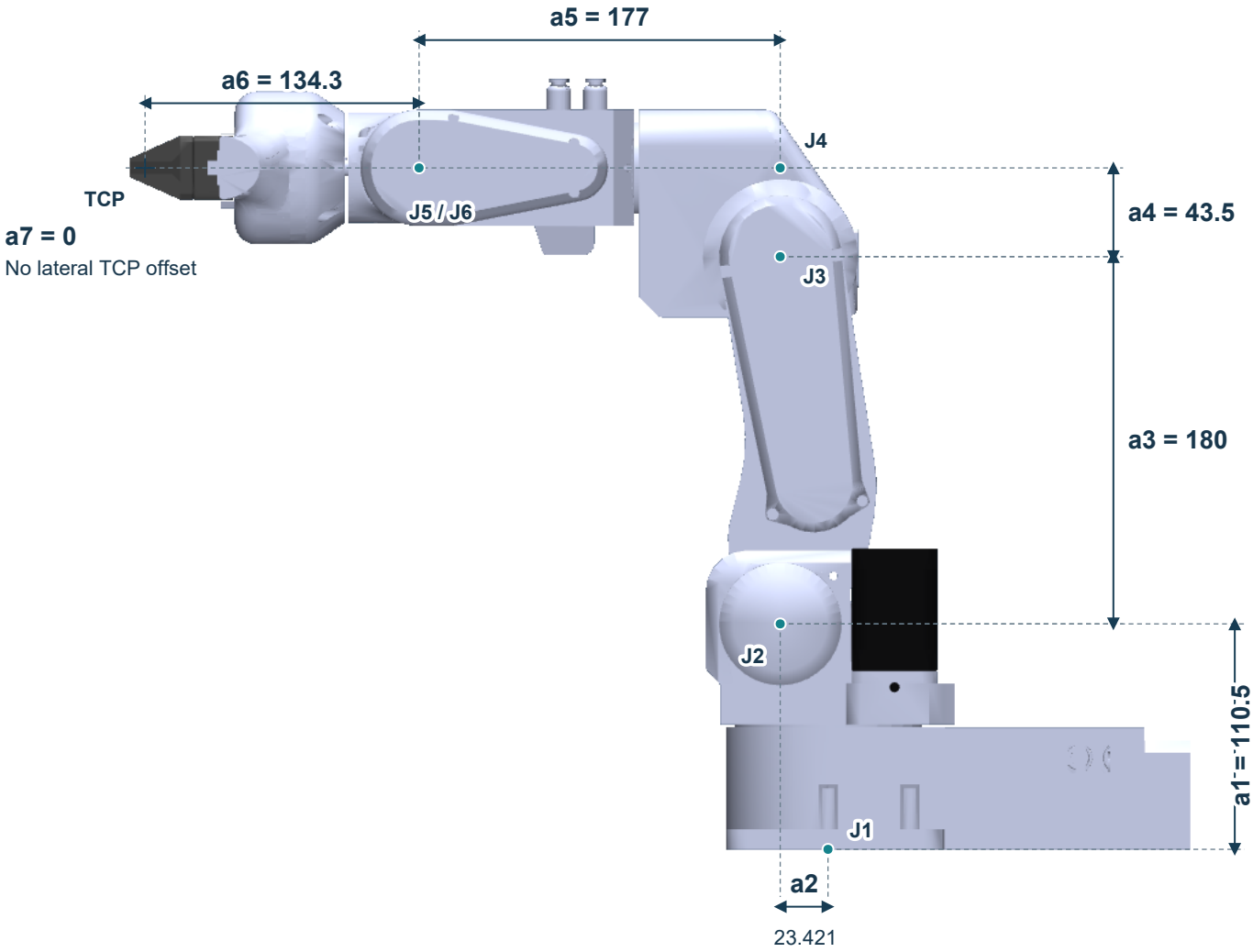
Source: parol6_msg_tpu_glue.urdf + scene_msg_tpu_glue.xml | Full transforms, hashes and checks: msg_tpu_glue.json

All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

PAROL6 / SSG48 / Solid jaws

Kinematic review | Standard DH to the configured TCP

Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.421	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = 0	-a6 = -134.3

SSG48 jaw variants share this operational TCP; it is not a measured grasp center.
Solid jaws are about 23.4 mm shorter than Fin-ray jaws; their TCP was retained.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.000818 mm / 0.000316 deg
Max ideal DH vs URDF: 0.002957 mm / 0.000764 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

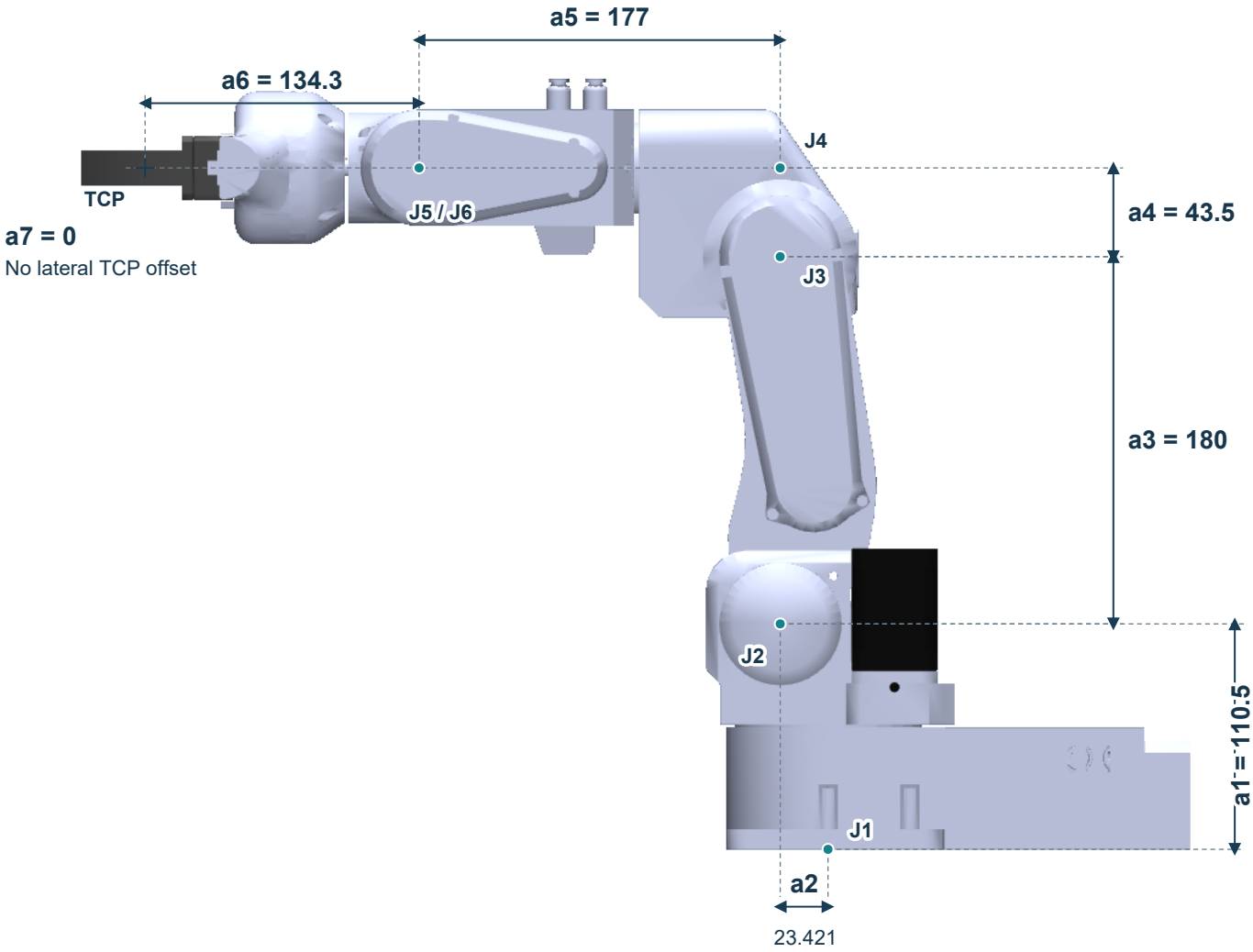
Source: parol6_ssg48.urdf + scene_ssg48.xml | Full transforms, hashes and checks: ssg48.json

All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

PAROL6 / SSG48 / Fin-ray jaws

Kinematic review | Standard DH to the configured TCP

Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.421	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = 0	-a6 = -134.3

SSG48 jaw variants share this operational TCP; it is not a measured grasp center.
Solid jaws are about 23.4 mm shorter than Fin-ray jaws; their TCP was retained.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.000818 mm / 0.000316 deg
Max ideal DH vs URDF: 0.002957 mm / 0.000764 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

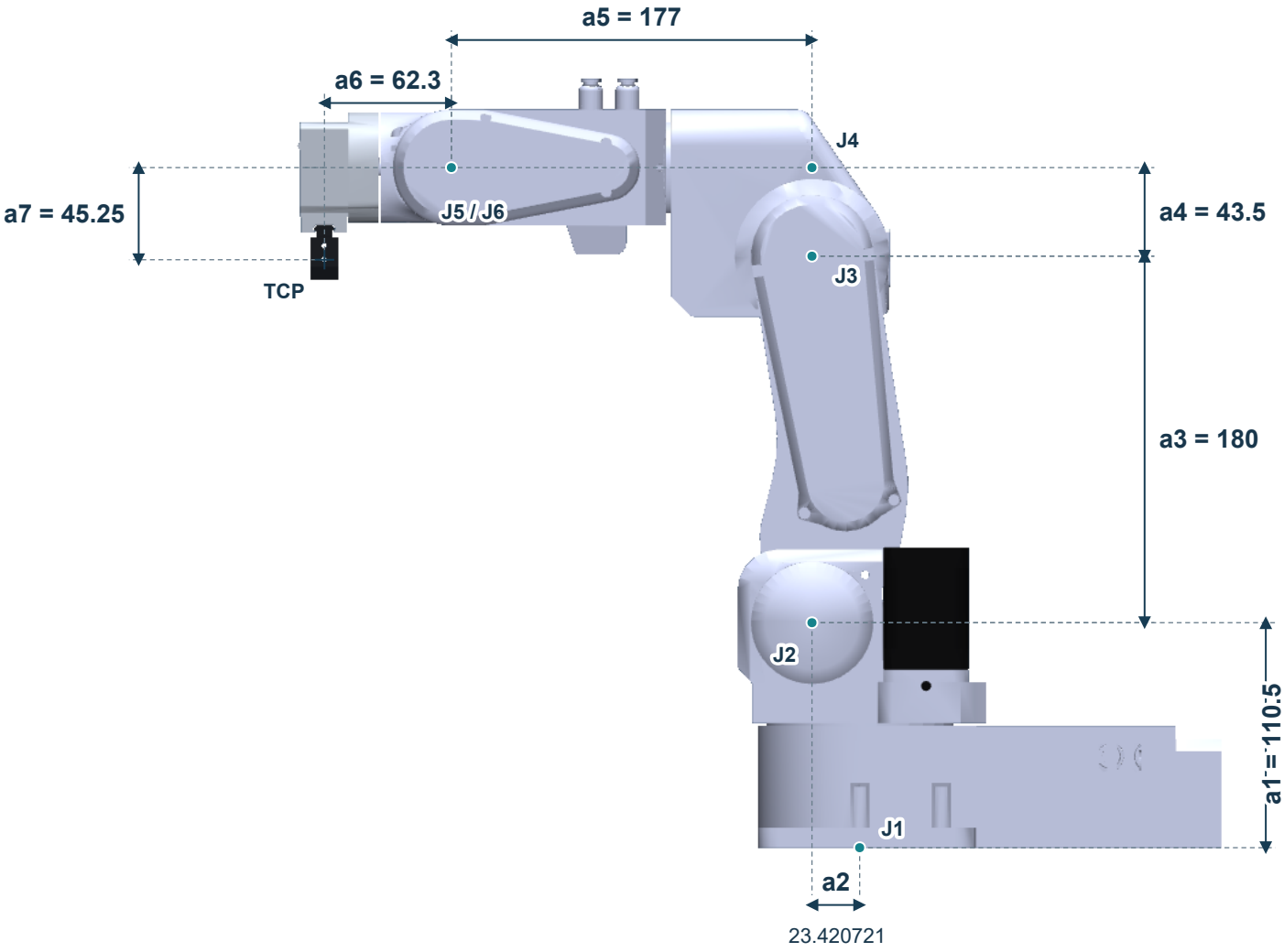
Source: parol6_ssg48_stock.urdf + scene_ssg48_stock.xml | Full transforms, hashes and checks: ssg48_stock.json

All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

PAROL6 / Pneumatic gripper

Kinematic review | Standard DH to the configured TCP

Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.420721	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = -45.25	-a6 = -62.3

Pneumatic TCP follows the legacy MHZ2-10D reference; it is not a newly measured grasp center.
Jaws close symmetrically through 3.5 mm each. Mass and finite actuator settings are estimates.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.002447 mm / 0.001026 deg
Max ideal DH vs URDF: 0.000000 mm / 0.000000 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

Source: parol6_pneumatic.urdf + scene_pneumatic.xml | Full transforms, hashes and checks: pneumatic.json

All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.